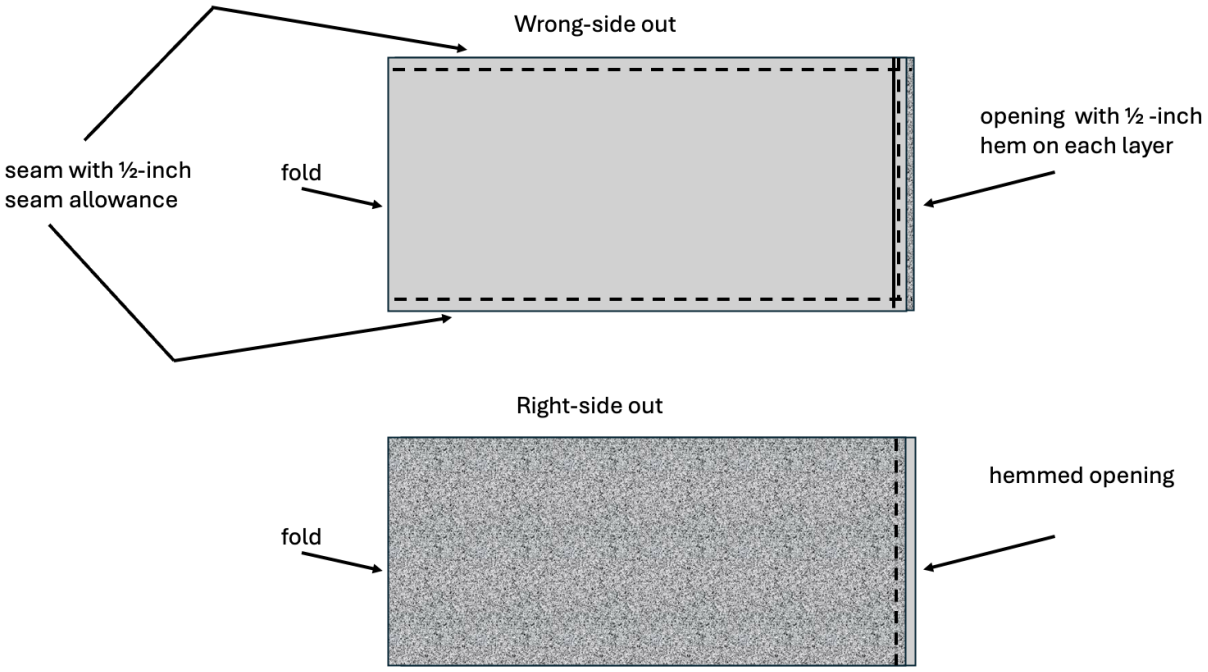
CSC 251 HOMEWORK TWO

YARDAGE CALCULATOR

THE ASSIGNMENT

You've adopted the most adorable puppy and plan to sew a rectangular pillowcase for a dog pillow. To make the pillowcase simple and to use your fabric efficiently, you'll fold the width of the fabric in half so that no seam will be required on the one side due to the fold. Opposite the fold, you'll turn each edge under ½ inch for a hem. You'll sew each of the other 2 sides (top and bottom) with a half-inch seam allowance, which, once the front and the back of the pillowcase are sewn together and turned right-side out, will be inside the pillowcase and not included in the final dimensions. You'd like to know how many yards of fabric you should purchase, depending on the length that you choose for the pillowcase, and the price of the cut of fabric.



In case you've never stepped foot into a fabric store, fabric comes in a few set widths, the most common of which is 44/45" wide. (The " means inches, by the way. Welcome to imperial units!) The long edges of the uncut fabric have a ½" border called a selvage that is printed with the manufacturer and style number of the fabric. This edge is usually trimmed off, hence only 44" of the 45" width is usable for your project. (This also gives them a little wiggle room in manufacturing.) The fabric is folded lengthwise and wrapped around a piece of cardboard that is approximately 24 inches long. This accommodates the folded width of 22.5" well and results in a "bolt" of fabric. When you purchase the fabric, you generally specify the desired **length to the nearest 1/8 of a yard**. At the fabric store, they will unwind the lose end of the bolt, measure that amount from the cut end of the fabric and cut the fabric across the width. So, when you ask for a yard of fabric, it is NOT a square yard (1 yard by 1 yard) but is a rectangular piece with a width of 1 yard (which is 36") and a width of 45".

So, if you purchase 1 yard of fabric for your dog's pillowcase, the finished pillowcase should measure:

21.5" wide: 45" minus two ½-inch selvages minus two ½-inch hems, folded in half

35" long: 1 yard = 36" minus two ½-inch seam allowances (one on each side)

If you desire the finished pillowcase to measure 36", you would need to buy 1 and 1/8 yards of fabric, written as 1-1/8 yards. (Of course, rather than wasting the extra 3.5" you would probably decide on a 39.5" finished-length pillowcase!)

Write a program to output the amount of fabric, to the nearest 1/8 of a yard, required to create the pillowcase with the specified finished length along with the fabric cut price without tax. Assume the width of the pillowcase is 21.5" as described above. Prompt the user with specific instructions regarding the values they should enter for both the finished pillowcase length to the nearest inch and the price of the fabric in units of dollars and cents per yard. **Implement input validation immediately after each input such that if they enter a nonsensical value for an input, you provide an error message and prompt them immediately to reenter that value until they enter a reasonable value**. If your user enters a length of 0 (which might be considered unreasonable, but we will allow it) your program should return a minimum fabric amount of 0 yards. You may assume that a pillowcase length over 3 yards is unreasonable. You are not expected to handle exceptions as when the wrong variable type is entered (string rather than a float or a float rather than an int) since we have not covered that yet in class.

Your output should appropriately indicate the singular "yard" for fabric cut to a length of a yard or less and plural "yards" for amounts over 1 yard. Amounts under 1 yard should be output as "3/8 yard", not "0-3/8 yard". Integer yard lengths should be output as "3 yards", not "3-0/8 yards". Fractions should be reduced to lowest form, so "2/8 yard" should be reduced to "1/4 yard".

Your program should contain **at least four** functions: a `main` function, two user-defined functions to assist in modularizing your code, and a test function. The test function should contain 3 tests to ensure that one of your 2 user-defined functions works as intended. Your `main` function should contain evidence of calling your test function, commented out so that it doesn't interfere with your final program execution. Organize your code wisely as this will count as a small percentage of your grade. You should use only constructs covered in class up through Lecture 04, therefore, no exception handling should be used. Built-in Python functions or functions from the math module may be used. Be sure your code is adequately documented and tested.

SAMPLE INPUT AND OUTPUT

User input is shown in bold for the sample runs below. Your user prompts do not need to match those shown but should be

Sample run #1:

```
Enter the length of the pillowcase in inches (0-108"): 36
Enter the price per yard: $4.99

Minimum fabric needed: 1-1/8 yards
Fabric cost: $5.61
```

Sample run #2:

```
Enter the length of the pillowcase in inches (0-108"): 13
Enter the price per yard: $11.99

Minimum fabric needed: 1/2 yard
Fabric cost: $6.00
```

Sample run #3:

```
Enter the length of the pillowcase in inches (0-108"): 1000
Invalid input for length. Re-enter the length in inches(0-108"): 54
Enter the price per yard: $9.59

Minimum fabric needed: 1-5/8 yards
Fabric cost: $15.58
```

Sample run #4:

```
Enter the length of the pillowcase in inches (0-108"): 54
Enter the price per yard: $-9.59
Invalid input for price. Re-enter the price per yard: $9.59

Minimum fabric needed: 1-5/8 yards
Fabric cost: $15.58
```

FILE SUBMISSION

You will submit a single source code file to Moodle using the naming format: assignment name, last name and first initial. Spelling and correctness will matter! Your documentation, use of functions, and style will also be checked.